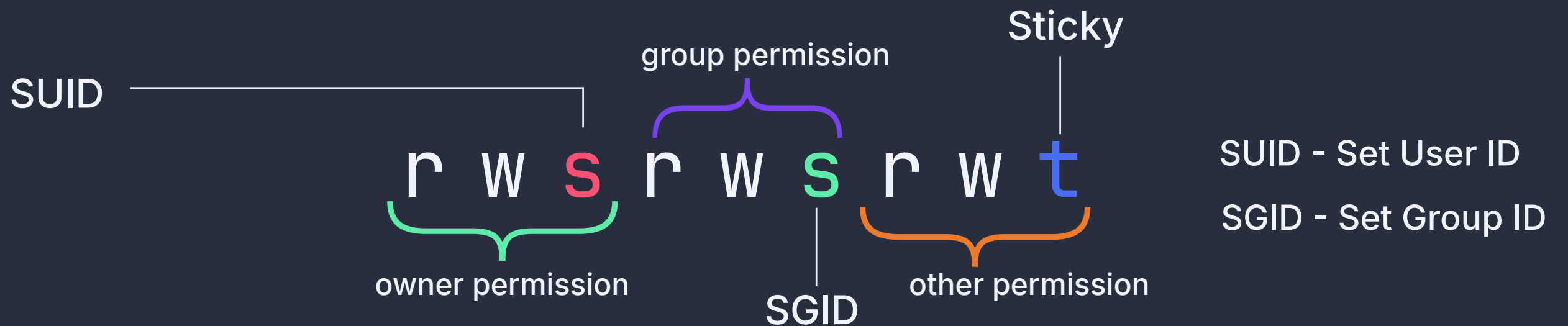


Special Linux Permissions



SUID bit

- Set on an executable file
- Allows the file to be executed with the permissions of the file owner
- Appears as 's' in owner execute positions
- Set with `chmod u+s filename`

SGID bit

- Set on an executable file or directory
- On files: Executed with permissions of the group owner
- On directories - New files inherit the directory's group ownership
- Set with: `chmod g+s filename`

Sticky bit

- Typically used on directories
- Prevents users from deleting files owned by other users
- Only file owner, directory owner or root can delete files
- Set with: `chmod +t directory`

Real World Examples

```
-rwsr-xr-x root root 59976 May 6 2022 /bin/password
```

Non-root users can run `passwd` as the root user to modify their own password in `/etc/passwd`

```
-rwxr-sr-x root crontab 39568 May 6 2022 /usr/bin/crontab
```

Users can run `crontab` with the `crontab` group's permissions to modify their cron jobs

```
drwxrwxrwt root root 36 Feb 2 17:21 /tmp
```

Files in `/tmp` can only be deleted by their owner, the directory owner (root), or root

Setting Up Permissions

Setting SUID

```
chmod u+s filename
```

```
chmod 4755 filename
```

Setting SGID

```
chmod g+s filename
```

```
chmod 2755 filename
```

Setting Sticky Bit

```
chmod +t directory
```

```
chmod 1777 directory
```

Setting Multiple

```
chmod ug+s, +t directory
```

```
chmod 7755 directory
```